



ING 3 IA --- Mobile Applications Practicum

Practical 2: The use of Flutter widgets for UI design

1) **Introduction** : the aim of this practical is to learn how to use essential components related to the building of Flutter mobile apps UI. Among the wide range of customizable widgets allowing you to create beautiful UI for your mobile apps, you will test some of them such as: Card, ListView, Icon, Specifically, you will create screens represented in figures 1, 2, 3 and 4.

Note: use the keyboard shortcut "***Ctrl+Space***" for auto-completion.

2) **App creation** :

1. Rename your Flutter project created in practical 1 to "***meal_planner***" and open it.
2. Replace the content of the "***main.dart***" file by the following code which is the basic code of a Flutter app :

```
import 'package:flutter/material.dart' ;  
void main () { runApp (MyApp() ); }  
class MyApp extends StatelessWidget {  
  @override  
  Widget build (BuildContext context) {  
    return MaterialApp (  
      title: 'My Meal Planner ',  
      theme: ???  
      home: HomeScreen (),  
    ); } }
```

3.To customize your app theme, replace ??? by a *"ThemeData"* widget. An example of customization could be as follows:

```
ThemeData (AppBarTheme: const AppBarTheme (  
    backgroundColor: Colors.black,  
    titleTextStyle: TextStyle (color: Colors.orange,  
    fontSize: 17, fontWeight: FontWeight.bold,),  
    iconTheme: IconThemeData (color: Colors.orange)),  
    ),
```



4.It is possible to remove the debug banner by adding the following line in the *"MaterialApp"* widget:

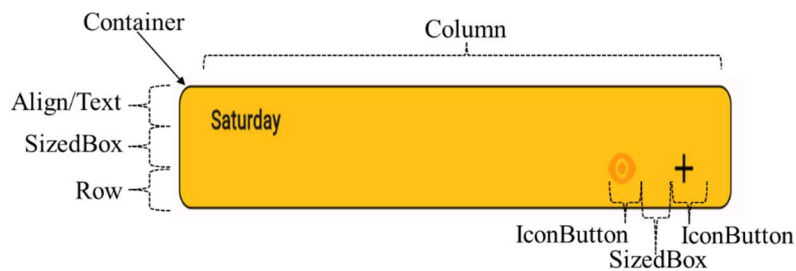
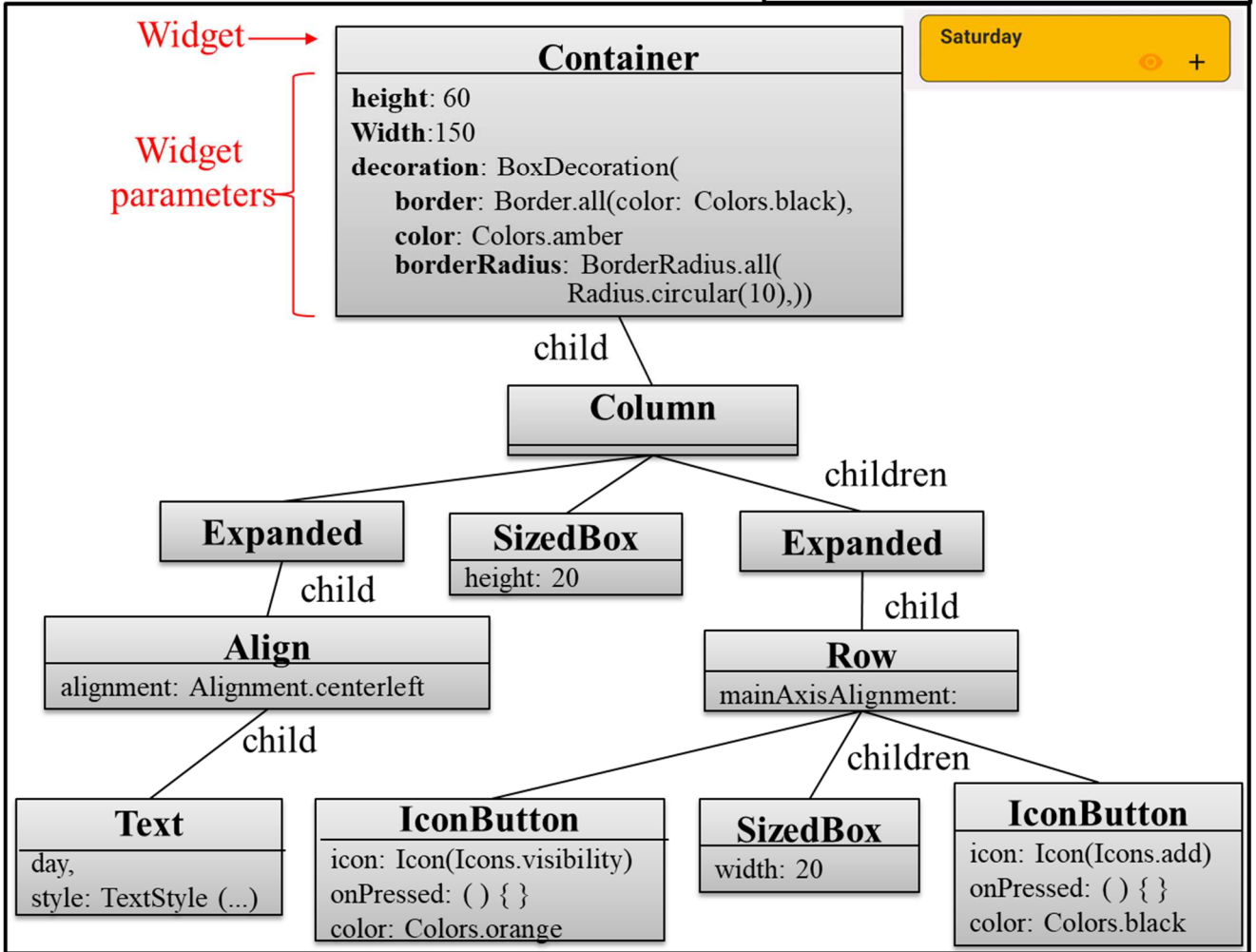
```
debugShowCheckedModeBanner: false,
```

3) **Trees widgets implementation:** based on the widgets seen in the class, create the following trees widgets:

a) **"WeekDaysCard" tree widgets:**

1. In the *"lib"* folder of your app, create a sub-folder named *"components"*. Then in the latter, add a new file named *"week_days_card.dart"*.
2. In *"week_days_card.dart"*, create a **Stateless** widget¹ (class) named *"WeekDaysCard"*. Then, add an attribute *"day"* **final** of **String** type and implement the *"WeekDaysCard"* constructor having as parameter the attribute *"day"*.
3. In the *"build"* method, construct the following tree widgets by specifying values to widget parameters:

¹ tapez st + CTRL + ESPACE et choisissez stateless widget.



b) "HomeScreen" tree widgets (Figure 1):

**WeekDaysCard
widget**

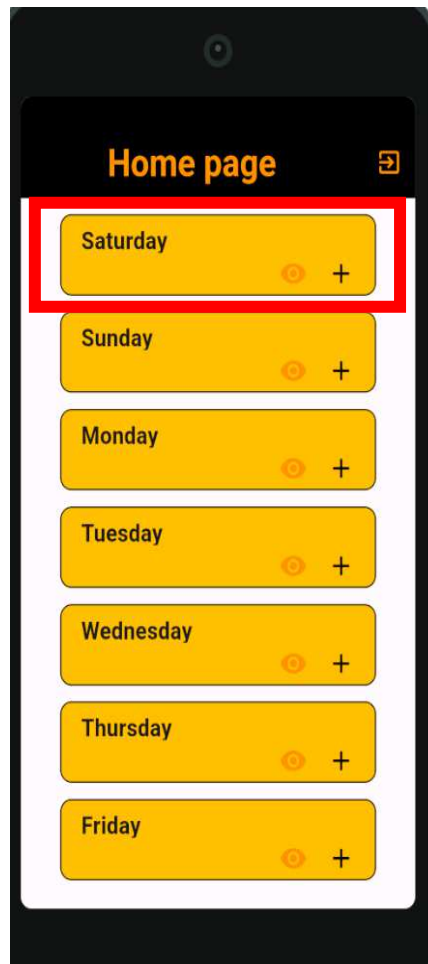
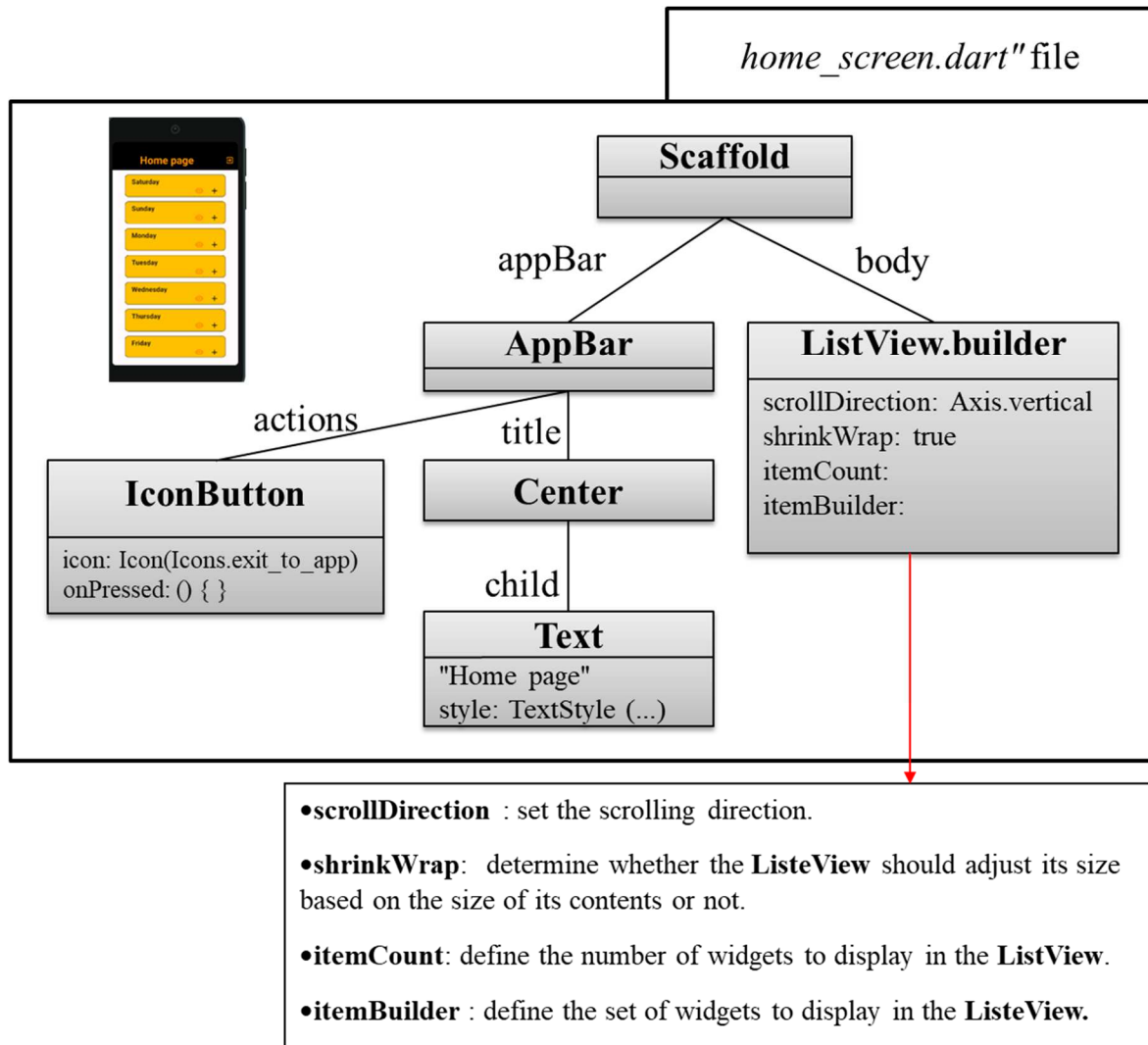


Figure 1: HomeScreen UI

1. In the *"lib"* folder of your app, create a sub-folder named *"screens"*. Then, in this latter, add a new file named *"home_screen.dart"*.
2. In *"home_screen.dart"*, create the class *"HomeScreen"* of type **Stateless** and add as a parameter a Dart list named *"week_days_list"* which is of type *String*. This list contains days of the week (Saturday, Sunday,..., etc.).
3. To produce the UI of the *Home page* (see Figure 1), implement the *"build"* method according to the bellow tree widgets:



- The "*itemBuilder*" property takes as value a function that returns the widget "*WeekDaysCard*" for a specific day among days defined in the Dart list of days (i.e. "*itemBuilder*" returns *WeekDaysCard* (*day* : *week_days_list*[*index*])).
- Run your application.

c) "LoginScreen" & "SignupScreen" tree widgets (Figures 2 & 3) :

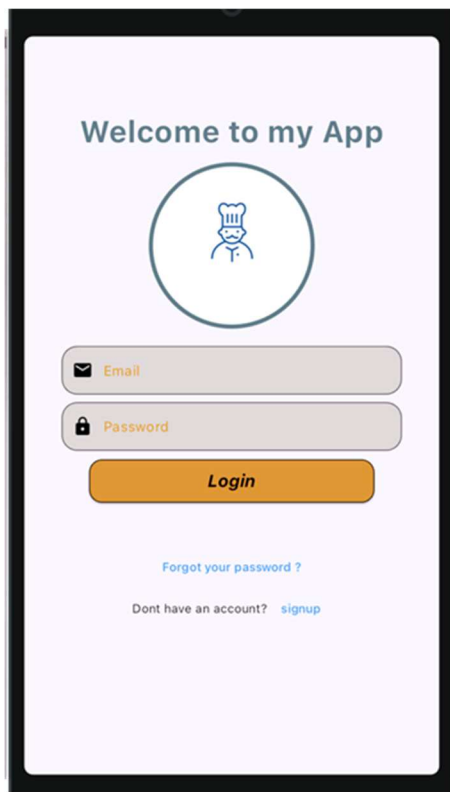


Figure 2: LoginScreen UI

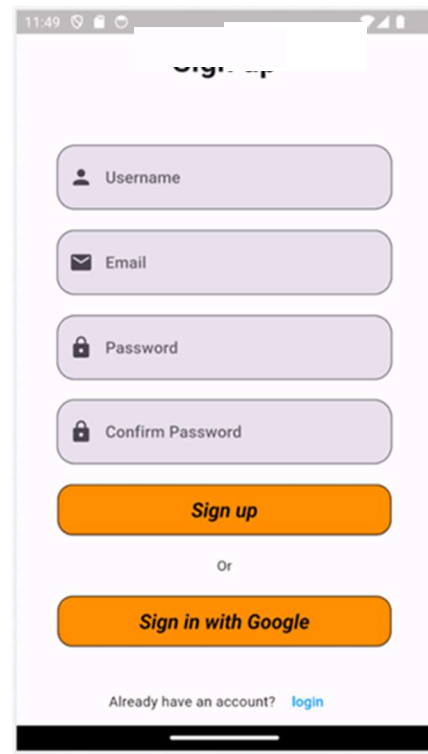


Figure 3: SignupScreen UI

1. To produce the UI of "LoginScreen" and "SignupScreen", it is necessary to use *TextField* widgets and *button* widgets ("*ElevatedButton*" & "*TextButton*"). From an optimization point of view (i.e. avoiding code redundancy), it is desirable to implement *TextField* widgets and *button* widgets in a way that they are reusable. To do that, in the "*components*" folder, add two new files named "*my_buttons.dart*" and "*my_textfield.dart*". In each file, add the Dart code corresponding, respectively, to the following trees widgets (each class is a **StatelessWidget**):

"my_buttons.dart" file

MyTextButton a **class** with two attributes:

- final String **buttonLabel**;
- final Function() **onPressedFct**;

MyElevatedButton a **class** with two attributes:

- final String **buttonLabel**;
- final Function() **onPressedFct**;



child

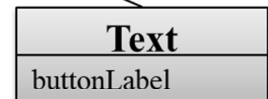


child



style

child



"my_textfield.dart" file

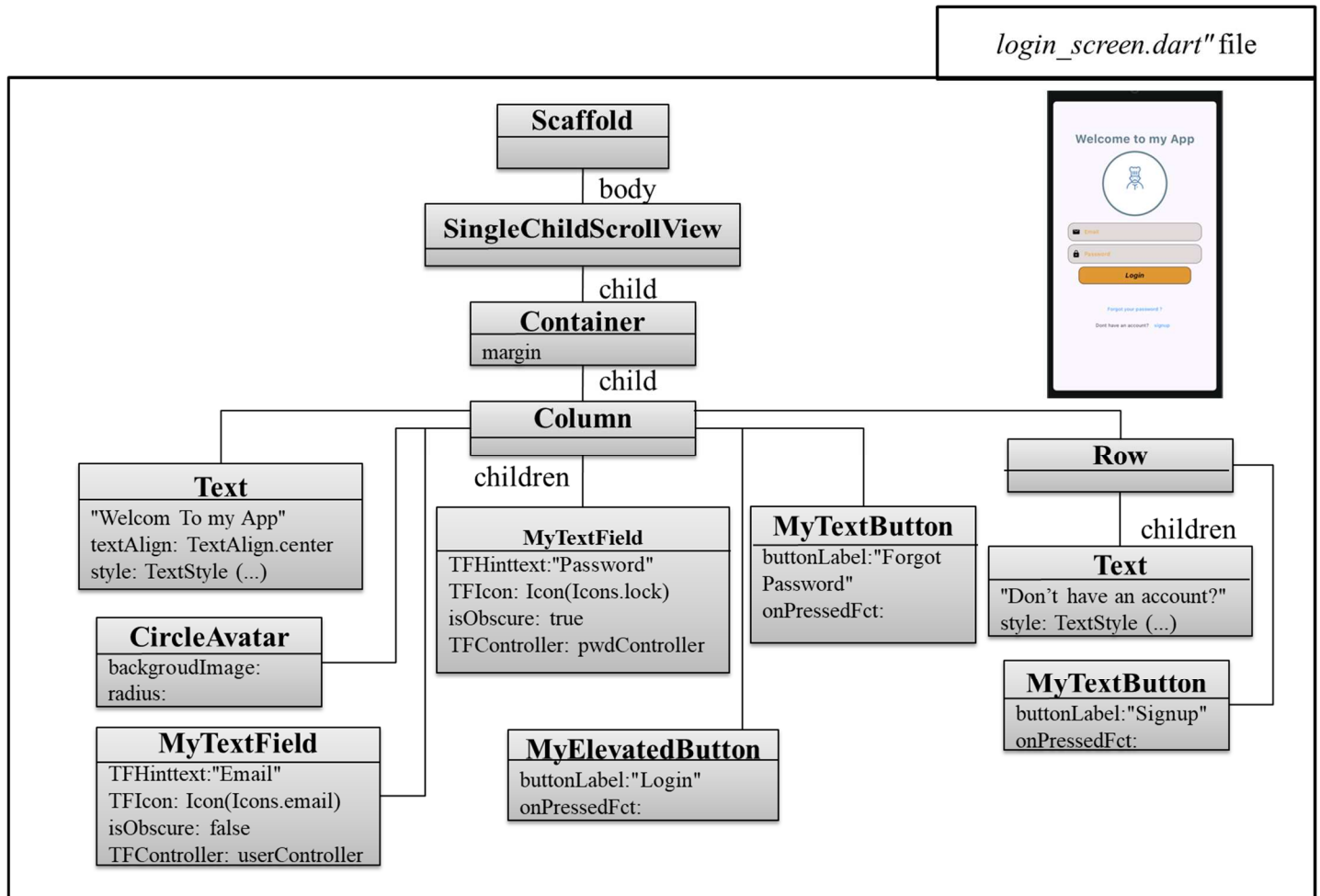
MyTextField a **class** with the following attributes:

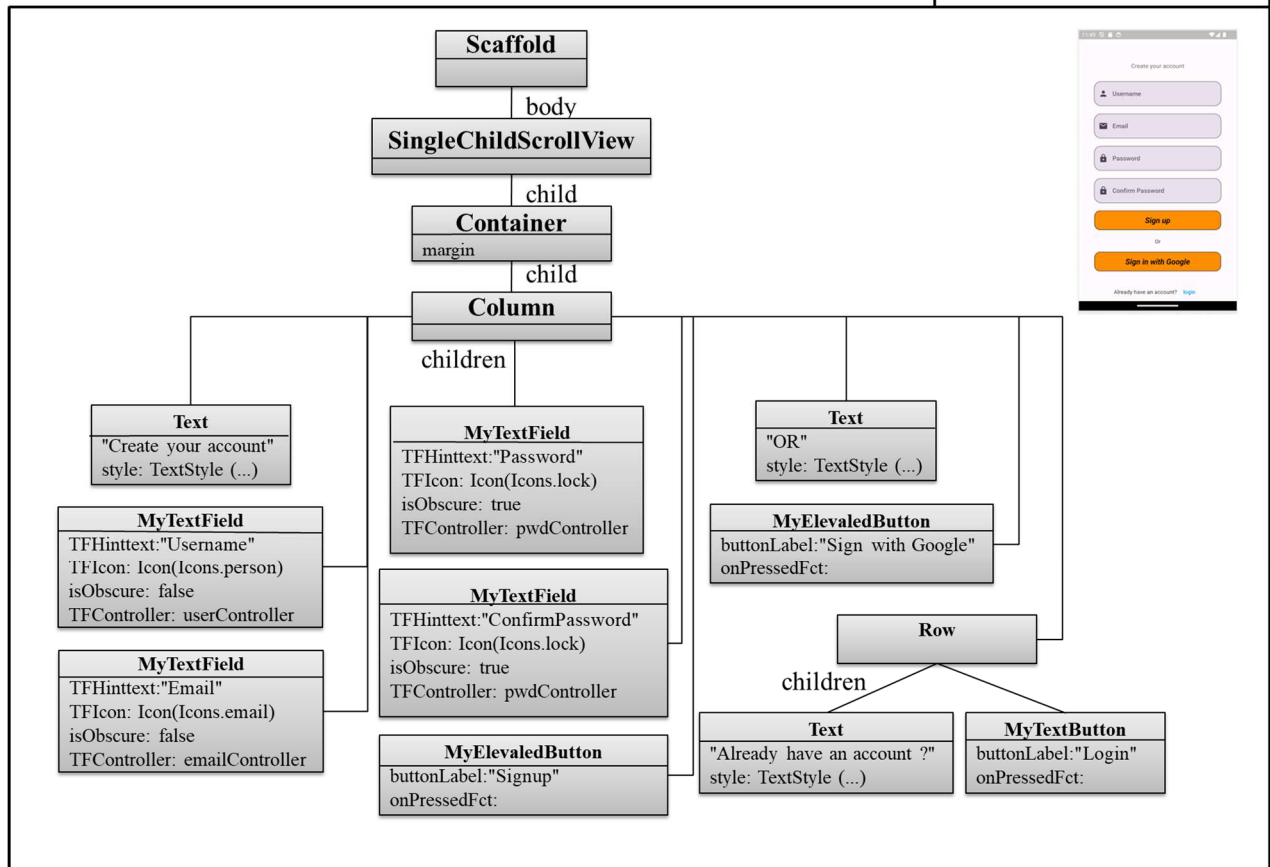
- final String **TFHintText**;
- final Icon **TFIcon**;
- final TextEditingController **TFController**;
- final bool **isObscure**;
- final String ? Function (String?) ? **TFValidator** ;

TextFormField

```
validator: TFValidator
controller: TFController
obscureText: isObscure
decoration: inputDecoration
(
  hintText: TFHintText
  border: OutlineInputBorder (borderRadius: BorderRadius.circular(18), )
  filled: true,
  prefixIcon: TFIcon
)
```

2. In the "screens" folder, add two new files named "login_screen.dart" and "signup_screen.dart". In each file, create a **Stateless** widget named "LoginScreen" and "SignupScreen", respectively. To implement the *build* methods, add the Dart code corresponding to the following trees widgets:





3. Set the *home* property of your *MaterialApp* to both *"LoginScreen()"* and *"SignupScreen()"*, then run your application.

d) "AddNewMealScreen", "MealsOfADayScreen" & "IngredientsOfAMealScreen" tree widgets (Figures 4):

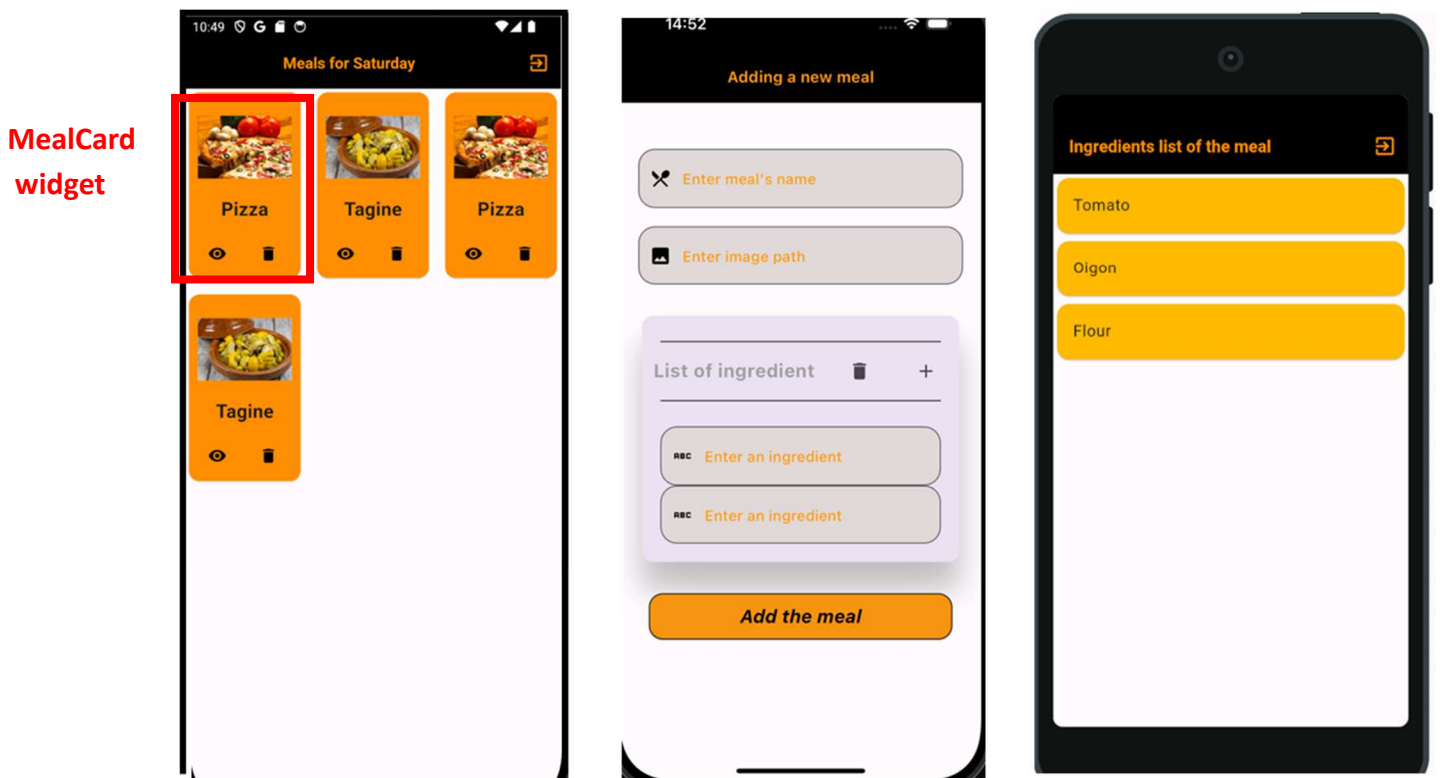


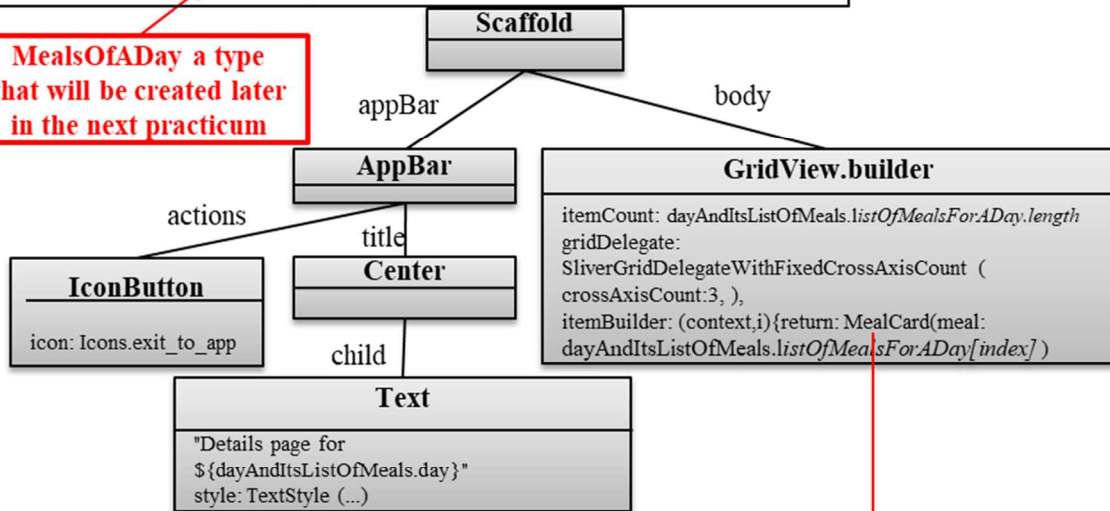
Figure 4: MealsOfADayScreen UI, AddNewMealScreen UI, & IngredientsOfAMealScreen UI

1. In the "screens" folder, create three new files named "meals_of_a_day_screen.dart", "add_new_meal_screen.dart" and "ingredients_ofa_meal_screen.dart".
2. In the "add_new_meal_screen.dart" and "meals_of_a_day_screen.dart" files, add a **Statefull** Widget named "AddNewMealScreen" and "MealsOfADayScreen", respectively. In the "ingredients_ofa_meal_screen.dart" file, add a **Stateless** Widget named "IngredientsOfAMealScreen".
3. To implement the "MealsOfADayScreen" widget, create a new file named "meal_card.dart" within the "components" folder (this file will contain the "MealCard" **Stateless** Widget). Then use the following tree widgets to implement the *build* methods.

"meals_ofa_day_screen.dart"

MealsOfADayScreen a **class** with the attribute:
• final **MealsOfADay** dayAndItsListOfMeals;

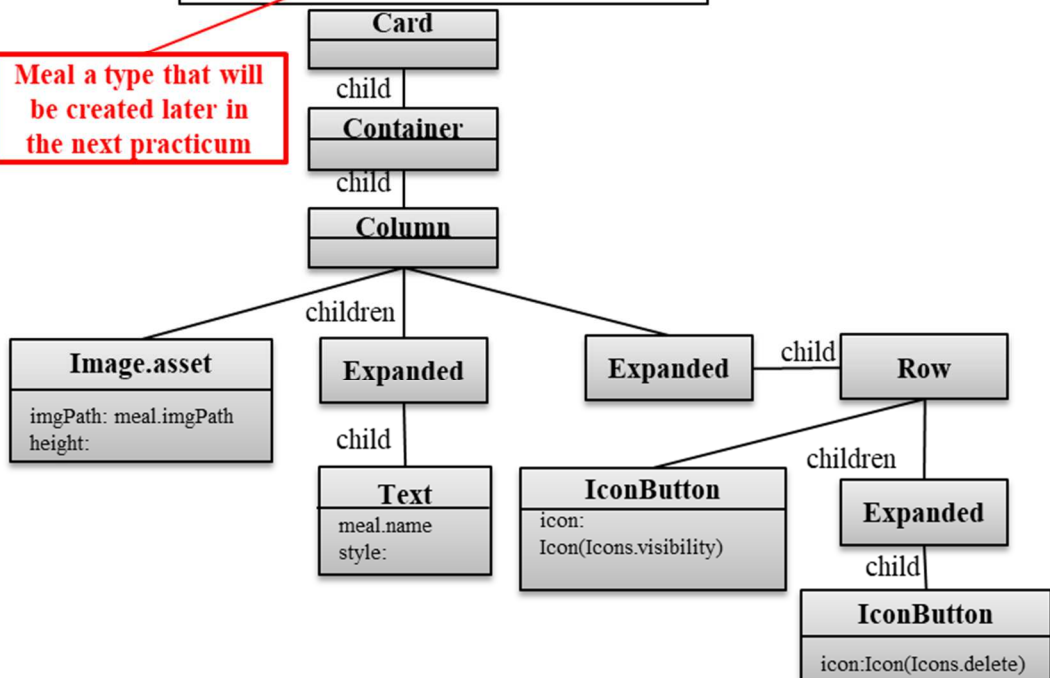
MealsOfADay a type
that will be created later
in the next practicum



"meal_card.dart"

MealCard a **class** with the attribute:
• final **Meal** meal;

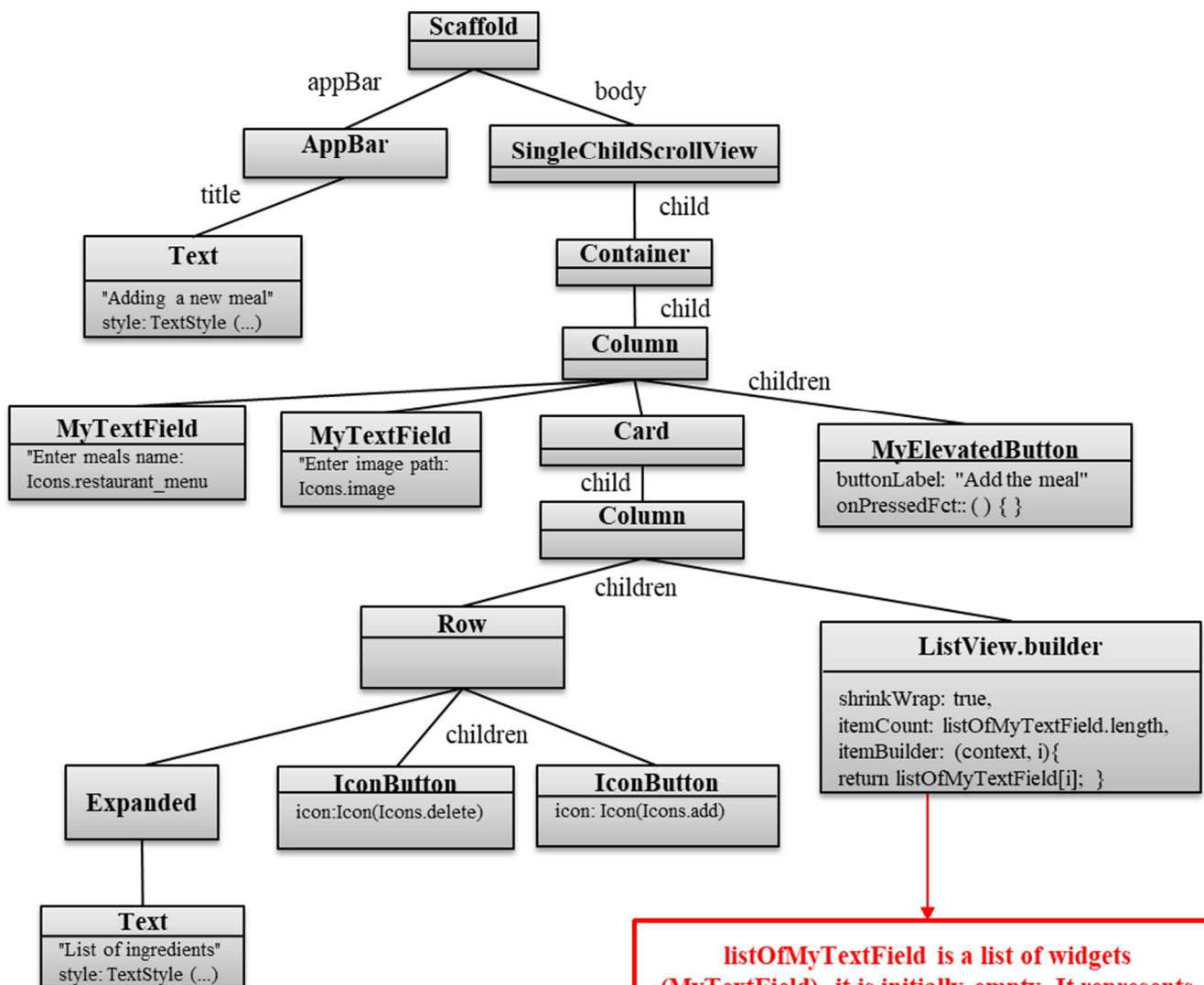
Meal a type
that will be created later
in the next practicum



4.The *build* method for "*AddNewMealScreen*" and "*IngredientsOfAMealScreen*" is implemented, respectively, according to the following trees widgets.

Note: pressing the icon button "*Add*" of the "*AddNewMealScreen*", will create a new *TextEditingController* and add it to a local variable named "*listOfTextField*". Pressing the icon button "*delete*" of the "*AddNewMealScreen*", will delete the last *MyTextField* in the "*listOfTextField*".

"add_new_meal_screen.dart" file



"ingredients_ofa_meal_screen.dart"

